Data Sources

Beyond Random Splits: Baselines, Validation Protocols and a Lag-Based Forecasting Model for Global Dengue Case Counts on OpenDengue

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This study uses two public datasets, four CSV files in total. The data are not included in this repository. Please download them from the official websites given below and place them as shown under Folder layout. The links are clickable.

Summary

#	File	Dataset and source	Used for
1	`national.csv`	OpenDengue v1.3 – global national data	Sections 3–5 (main analysis)
2	`Spatial_extract_V1_3.csv`	OpenDengue v1.3 – global highest spatial resolution data	Section 5.6 (sub-national transfer test)
3	`dengue_features_train.csv`	DengAI (DrivenData) – Training Data Features	Section 5.9 (external validation)
4	`dengue_labels_train.csv`	DengAI (DrivenData) – Training Data Labels	Section 5.9 (external validation)

Dataset 1: OpenDengue (version 1.3)

A global database of publicly reported dengue case counts. No login is needed. Licence: CC BY-SA.

Resource	Link
Download page	https://opendengue.org/data.html
GitHub (all versions)	https://github.com/OpenDengue/master-repo
Data paper (*Scientific Data*, 2024)	https://doi.org/10.1038/s41597-024-03120-7
Dataset archive (Figshare)	https://doi.org/10.6084/m9.figshare.24259573

How to download. Open the download page and go to the *Download global data* section. Use these two buttons:

Button on the website	Save the file as	Used for
Download global national data	`national.csv`	Main analysis (Sections 3–5)
Download global highest spatial resolution data	`Spatial_extract_V1_3.csv`	Sub-national test (Section 5.6)

The button for global highest temporal resolution data is not needed; it is not used in this study.

Check your download. The national extract used here has about 29,873 rows, 129 countries and territories, and covers 1924–2025. If yours differ, use version 1.3.

Dataset 2: DengAI (DrivenData competition)

Weekly dengue cases and environmental features for San Juan (Puerto Rico) and Iquitos (Peru). Used only for the external validation in Section 5.9. A free DrivenData account is needed: join the competition, open the *Data* tab and download the two training files.

Resource	Link
Competition page	https://www.drivendata.org/competitions/44/dengai-predicting-disease-spread/

File on the website	Save the file as
Training Data Features	`dengue_features_train.csv`
Training Data Labels	`dengue_labels_train.csv`

Test Data Features and Submission Format are not used. Please read DrivenData's terms of data use before reusing the data; the DengAI files are not included in the code repository.

Folder layout

```
project/
|-- dengue_revised.ipynb
|-- national.csv
|-- Spatial_extract_V1_3.csv
`-- data/
    `-- dengai/
        |-- dengue_features_train.csv
        `-- dengue_labels_train.csv
```

The notebook reads its paths from environment variables, which can be changed:

Variable	Default	Meaning
`DENGUE_CSV`	`national.csv`	OpenDengue national extract
`OUT_DIR`	`results`	Output folder
`DENGAI_DIR`	`data/dengai`	Folder with the two DengAI files

The spatial extract file name is set by `SPATIAL_RAW_CSV` in the notebook; it is only needed the first time, to build the cached sub-national file.

How to cite the data

OpenDengue: Clarke J, Lim A, Gupte P, Pigott DM, van Panhuis WG, Brady OJ. A global dataset of publicly available dengue case count data. *Scientific Data* 11, 296 (2024). https://doi.org/10.1038/s41597-024-03120-7

DengAI: DrivenData. DengAI: Predicting Disease Spread. https://www.drivendata.org/competitions/44/dengai-predicting-disease-spread/